AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2016

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 9, 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

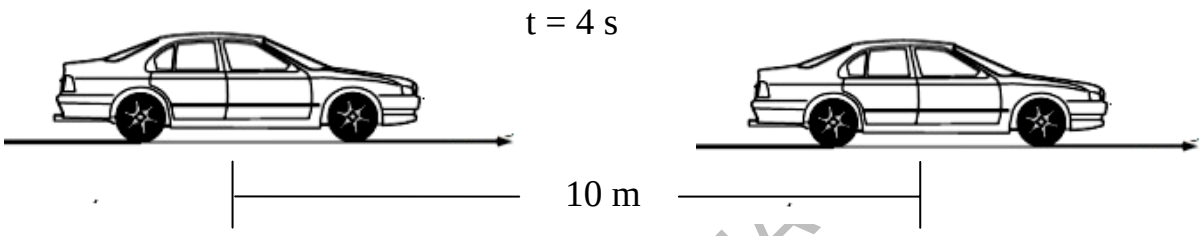
5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Match the physical quantities in column **A** with the SI units in column **B** using arrows (⟶).

Column A	Column B
Intensity of light	Ampere
Temperature	Mole
Amount of substance	Kelvin
	Candela

Q.2. (Total 3 Marks)



In the above diagram a car starts from rest and after time t = 4 s covers a distance of 10 m. Calculate the acceleration of the car.

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Q.3. (Total 3 Marks)

a. Describe the force which causes a ball to return back to the ground in TWO points. (2 Marks)



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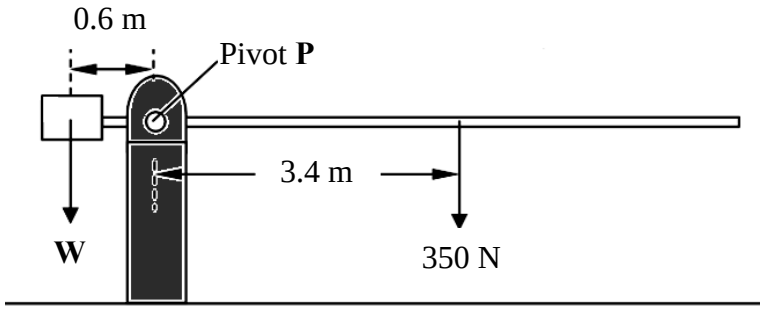
b. What is the difference between mass and weight? (1 Mark)

Mass	Weight

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Q.4. (Total 3 Marks)

An automatic barrier is shown in the given diagram. The barrier arm is balanced by the block of weight **W**.



a. Using the data given in the above diagram, calculate the weight **W** of the block. (2 Marks)

b. State the principle which is applied to solve the above problem. (1 Mark)

_____

_____

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Q.5. (Total 3 Marks)

Two identical balls of masses $\mathbf{m_1}$ and $\mathbf{m_2}$ whereas ($\mathbf{m_1 = m_2}$) are separated by a distance of **r**. If the distance between the two balls is doubled, form an equation proving that the gravitational force $\mathbf{F_G}$ between them will be decreased one-fourth of the initial force.

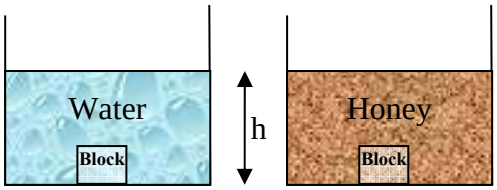
Q.6. (Total 3 Marks)

A car of mass **m** starts moving from rest and covers a distance **S**. If the final velocity is **v** at time **t**, then derive an expression of kinetic energy for the car.

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Q.7. (Total 3 Marks)

Given below are two identical beakers containing two different liquids of the same volume.
An iron block of equal size is placed in each of the beakers.



A **B**

a. Define density. (1 Mark)

_____

_____

b. In which above given figure (**A** or **B**) the pressure on the block is greater? Justify your answer with the help of an equation. (2 Marks)

_____

_____

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Q.8. (Total 4 Marks)

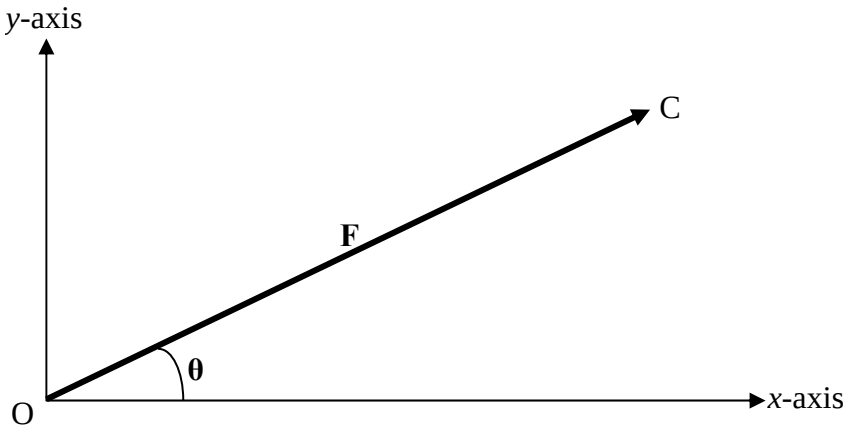
If the celsius thermometer shows the temperature of air as 30 ^{o}C at a certain day, then calculate the temperature of air in kelvin and fahrenheit.

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Q.9. (Total 5 Marks)

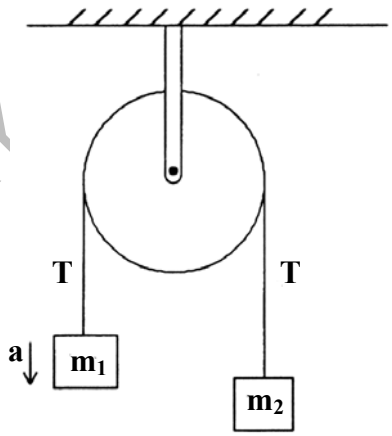
EITHER

a. The given vector **F** is making an angle **θ** with the positive *x*-axis. Resolve the given vector **F** into its rectangular components. Also mention a formula for finding out the magnitude of a single vector and direction from its resolved components.



OR

b. Two masses $\mathbf{m_1}$ and $\mathbf{m_2}$ are attached to the ends of a string that passes over a frictionless pulley.



If $\mathbf{m_1}$ is greater than $\mathbf{m_2}$, then determine the acceleration of the two masses when released. Show your answer in terms of $\mathbf{m_1}$, $\mathbf{m_2}$ and **g**. (Where **g** = acceleration due to gravity)

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Q.10. (Total 5 Marks)

EITHER

a. A block of ice having volume 10 m^3 hits the ground with the velocity of 2 m/s.

Find its kinetic energy in joules taking the density of ice block as 920 kg/m^3.

OR

b. Find the pressure due to the fluid at a depth of 10 cm in still water and in mercury.

(**Note:** The density of water and mercury is 1000 kg/m^3 and 13.6 g/cm^3 respectively.)

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Q.11. (Total 5 Marks)

EITHER

a. Define the term evaporation and write any FOUR factors that affect it.

OR

b. Radiation is a method of heat transfer whereby heat energy is transmitted from one hot object to another in the form of infra-red. Write any FIVE applications of radiation.

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END OF PAPER

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Please use this page for rough work

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